

2.5.24

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Question: If $|\vec{a}| = 2$, $|\vec{b}| = 7$ and $\vec{a} \times \vec{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, find the angle between $\vec{a}$ and $\vec{b}$

Solution:

$$|\vec{a}| = 2, \quad |\vec{b}| = 7, \quad \vec{a} \times \vec{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$$

Let the directions of $\vec{a}$ and $\vec{b}$ be such that

$$\vec{a} = (a_1, a_2, a_3)^\top \quad (1)$$

$$\vec{b} = (b_1, b_2, b_3)^\top \quad (2)$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (3)$$

$$\vec{a} \times \vec{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}^\top \quad (4)$$

Given:

$$\vec{a} \times \vec{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}^\top = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix}^\top \quad (5)$$

$$|\vec{a} \times \vec{b}| = \sqrt{3^2 + 2^2 + 6^2} = \sqrt{9 + 4 + 36} = \sqrt{49} = 7 \quad (6)$$

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta \quad (7)$$

where the angle between $\vec{a}$ and $\vec{b}$ is θ

$$7 = 2 \times 7 \sin \theta \Rightarrow 7 = 14 \sin \theta \Rightarrow \sin \theta = \frac{1}{2} \quad (8)$$

Therefore, the angle between $\vec{a}$ and $\vec{b}$ is $\theta = 30^\circ$

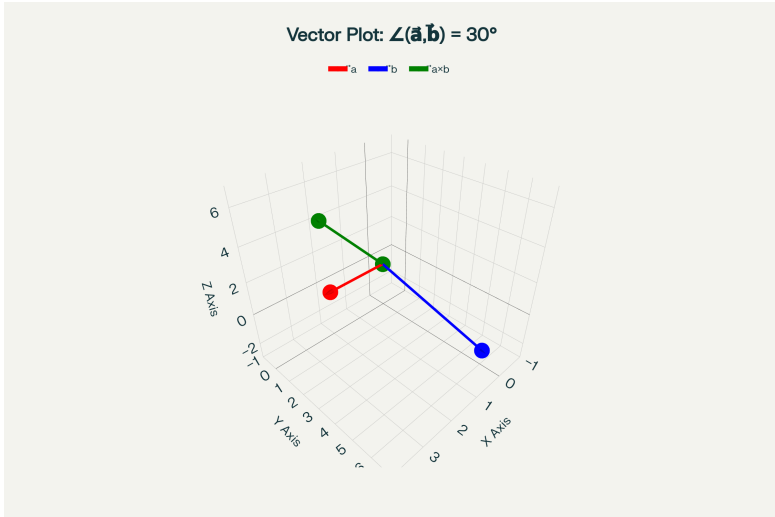


Fig. 0: A vector diagram showing vectors $\vec{a}$, $\vec{b}$, and their cross product with a 30° angle